PRAISING HARRIS

SOFTWARE ENGINEERING / COMPILERS & ASTs / DISTRIBUTED SYSTEMS / OPEN SOURCE

9597691821 | praisingharris@gmail.com | github.com/Pyhroff | medium.com/@praisingharris
linkedin.com/in/praising-harris-8a3320330 | tryhackme.com/p/Pyhroff

2 Merged Tier-1 PRs SQLGlot & LiteLLM · **16 Upstream PRs** Graphene, SQLFluff, MCP · **30/30 Raft Scenarios** Kronos-KV split-brain passed · **100 s → 10 s** profiled latency fix

EDUCATION

Indian Institute of Information Technology, Kottayam 2024 – 2028 (expected)
B.Tech Computer Science & Engineering (Cybersecurity) | CGPA 7.9
6-Month Co-op Availability: Nov 14, 2026 – Summer 2026 | University NOC Approved & Academic Credits Eligible
Coursework: Distributed Systems, Operating Systems, Compilers, Algorithms, Network Security | DSA: NeetCode 150 in progress

PROJECTS

Upstream Systems & Compiler Engineering: SQLGlot, Graphene, SQLFluff github.com/Pyhroff
SQLGlot (Merged #8780) · Graphene (#1600) · SQLFluff (#8899–#8902) · Anthropic MCP SDK · LiteLLM

- Toby Mao's SQLGlot (Merged #8780): Fixed AST node mutation in lambda-transpilation engine (_replace_lambda) by isolating DataType deep copies across casts, eliminating silent query corruption in multi-argument lambdas with zero allocation overhead.
- graphql-python/graphene (PR #1600) & SQLFluff: Authored lazy type resolution for Connection.Meta.node (#1600) resolving circular import deadlocks in Relay schemas; authored 4 compiler grammar fixes in SQLFluff (#8899–#8902) across StarRocks, ANSI, and Doris dialects.
- Distributed & Protocol Systems: Shipped merged engine fix in BerriAI/litellm (PR #47091: Cohere stop mapping), authored RFC 6749/9110 OAuth and header whitespace compliance in Anthropic MCP SDK (#3764, #3765, 175 tests), and 5 runtime fixes in crewAI (MCP pagination #8240, Redis URL password unquote #8244).

OSIRIS: Multi-Provider Distributed Agent Backend
Python · FastAPI · SQLite · Groq / Gemini / Ollama / OpenAI · AsyncIO · Docker

- Built an asynchronous provider router across 8 LLM backends with priority ordering, automatic failover, per-provider cooldowns, and SQLite persistence with embedding recall; integrated Gmail, Calendar, Drive, and GitHub APIs behind FastAPI.
- Profiled a 100 s local-model call to Ollama's OpenAI-compatible endpoint ignoring latency settings, wrote a native adapter cutting plain tasks to 10 s; 100 tests pass in CI.

Kronos-KV: Distributed Storage Engine & Raft Consensus github.com/Pyhroff/kronos-kv
Python (Zero-Dep stdlib) · Bitcask LSM · Raft Consensus · Consistent Hashing · 8/8 pytest pass

- Storage Engine: Built Bitcask LSM engine with CRC32 binary WAL, immutable SSTables, and in-memory KeyDir achieving O(1) point reads at 0.11ms/read; merge-sort compaction reclaimed 84.29% disk space (870KB → 136KB) with Kirsch-Mitzenmacher double-hashing Bloom filters (0.19% FPR).
- Distributed Consensus & Sharding: Implemented 5-node Raft protocol (§5.2 election safety, §5.3 log replication, §5.4 commit rule) with 100% split-brain prevention across 30 partition scenarios in 7.24ms; built 150-vnode consistent hash ring limiting key remapping to 9.47% on node additions (8 → 9 shards).

SOC PARALLAX: Event Ingestion & Graph Backend
Python · FastAPI · PostgreSQL · Neo4j · Next.js · Docker Compose

- Designed an ingest pipeline normalizing Sysmon / EVTX, Windows Event, JSON / ECS and CSV logs into one schema, storing events and baselines in PostgreSQL, and serving REST APIs to a Next.js dashboard; entire stack orchestrated in Docker Compose with 100% pytest test coverage.

SKILLS

Languages Python, JavaScript, SQL, Bash, C/C++

Distributed & Systems Raft Consensus, Bitcask LSM, Consistent Hashing, FastAPI, AsyncIO, SQLite, PostgreSQL, Neo4j, Docker

Engineering & Testing AST manipulation, compiler grammars, pytest, GitHub Actions, Linux, CI/CD pipelines

ACHIEVEMENTS & HACKATHONS

Hackathons **PSBs Hackathon 2026 (PhantomGrid: FastAPI backend for behavioural-biometric authentication with replay defence and SHA-256 hash-chained audit log); USAII Global AI 2026 (Lucid).**

Other TryHackMe top 2% globally, 177 rooms. Six technical articles on Medium (medium.com/@praisingharris).